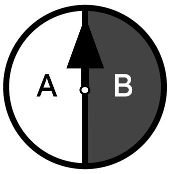
Name: ______________________

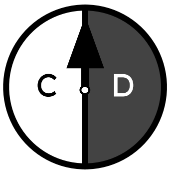
Worksheet

Using an outcome tree to display outcomes for more than two events

Task A: Outcome trees for multi-stage trials



spinner 1

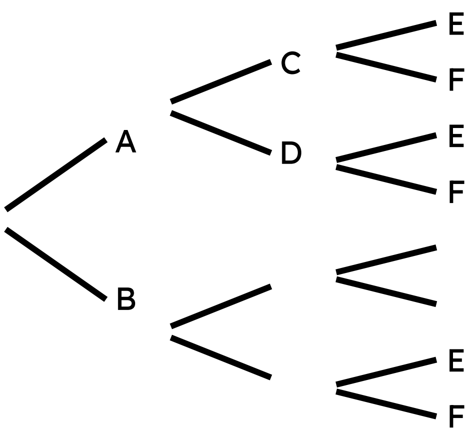


spinner 2

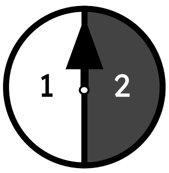
spinner 3

1) Sofia will spin spinner 1, 2, and 3 once each.

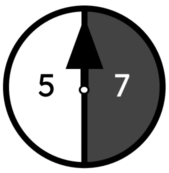
a) Complete this **outcome tree** and sample space.



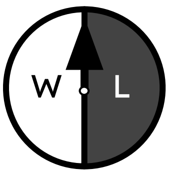
b) The event that at least one spinner lands on a vowel has how many outcomes?



spinner 1



spinner 2

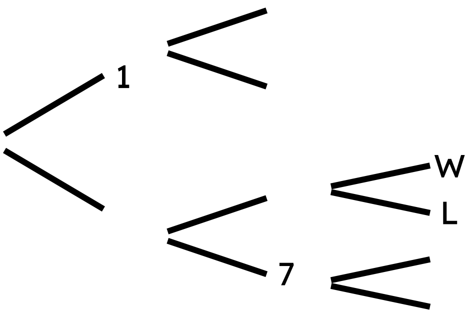


spinner 3

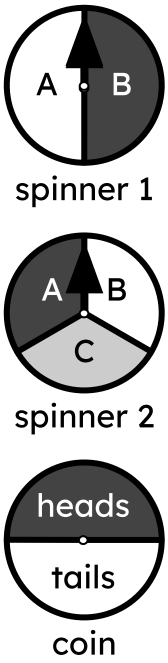
2) Aisha plays a game.
Aisha will spin spinners 1, 2, and 3 once each.

If spinner 3 lands on W, then Aisha wins the number of sweets equal to the **sum** of the numbers on spinners 1 and 2.

Complete this **outcome tree** and sample space for Aisha's game.



1 5 W	wins 6
2 5 W	wins 7
2 5 L	no win



3) Sam will spin spinners 1 and 2 once each, and then flip a coin.

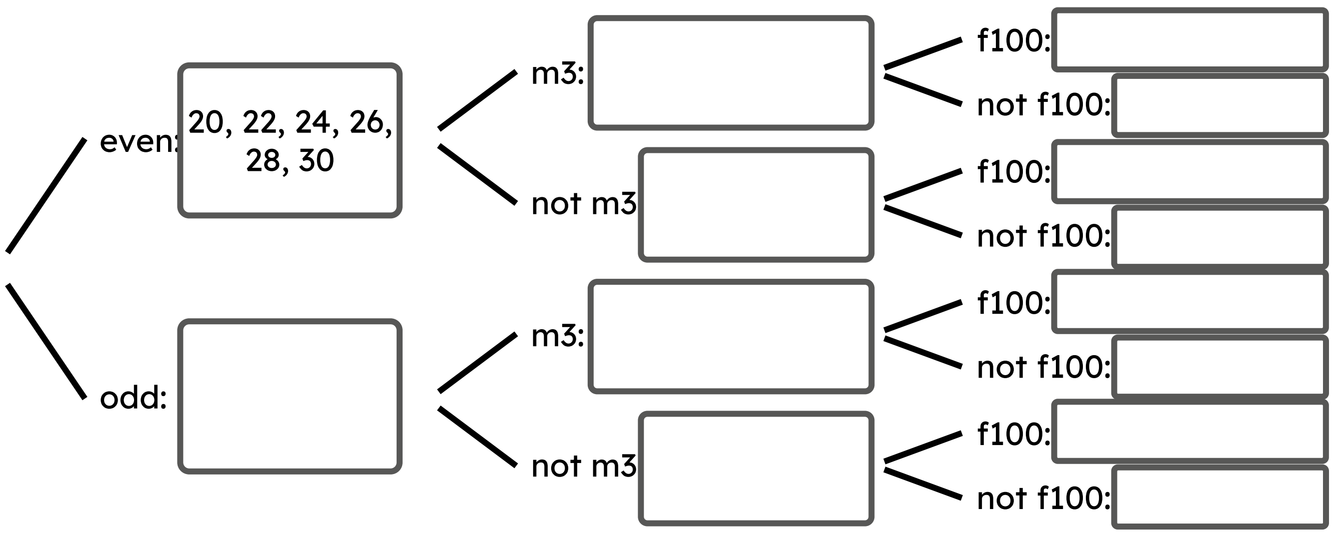
a) Draw the **outcome tree** for this three-stage trial. Stage 1 is spinner 1, stage 2 is spinner 2, and stage 3 is the coin.

b) Modify this **outcome tree** to satisfy this four-stage trial: “if two spinners land on the same letter and the coin lands heads, then flip another coin”.

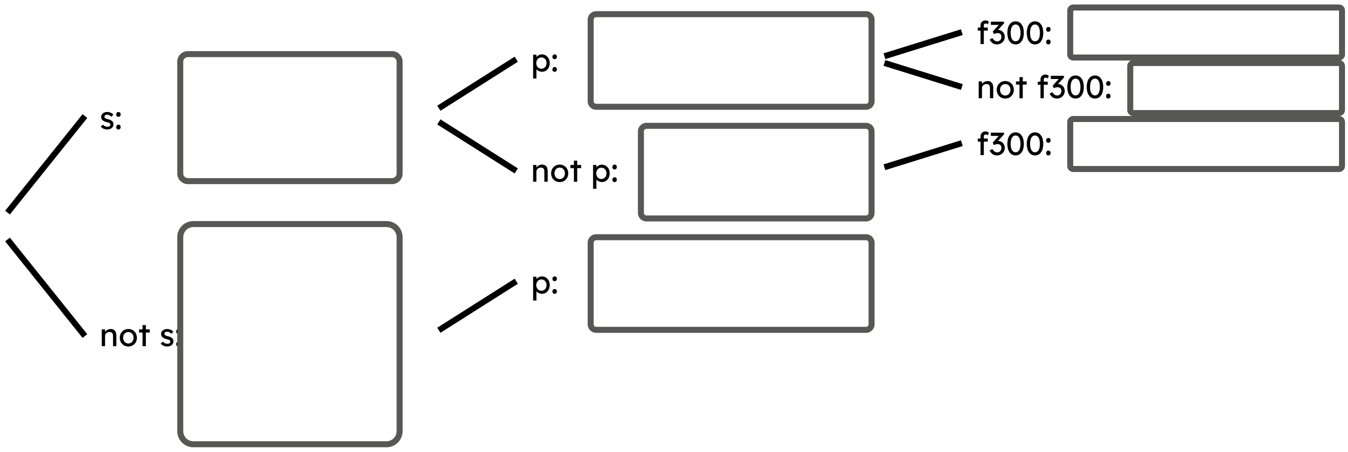
Task B: Outcome trees from multiple events

1) Sam will spin a spinner with the integers from 20 – 30 on it.
Sam considers the events that the spinner lands on an even number, a multiple of 3 (m3) and a factor of 100 (f100).

Fill in the blanks for this **outcome tree** with outcomes from this spinner.

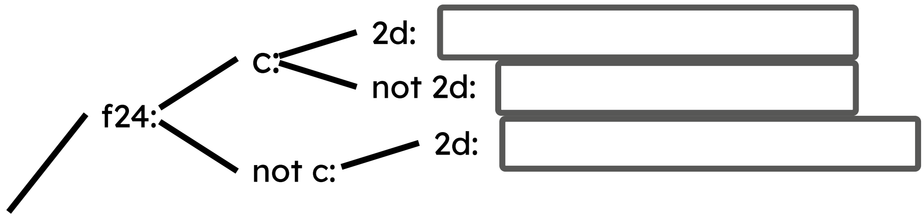


2) Lucas will spin a spinner with the integers from 15 – 25 on it.
Lucas considers the events that the spinner lands on a square number (s), a prime number (p), and a factor of 300 (f300).
Fill in the blanks for and complete this **outcome tree**.



3) Jun will spin a spinner with the integers from 0 to 15 on it.
Jun considers the events that the spinner lands on a factor of 24 (f24), and a cube number (c), and a two-digit number (2d).

a) Complete the partial **outcome tree** and list the outcomes at the end of each branch.



b) How many outcomes match the events that Jun is looking out for?

4) Sofia has a spinner with the following outcomes:
{1, 3, 4, 5, 8, 10, 12, 20, 25, 27, 36}

a) Construct and fill in a four-layer **outcome tree** for the outcomes of Sofia's spinner.

b) How many outcomes that are even numbers are also both square and triangular?